BANJI OYEWOLE

STAFF SOFTWARE ENGINEER

www.banj.io | banji@banj.io | San Francisco, CA

SUMMARY

Staff Software Engineer — Apps people use to see, control, and trust real-time hardware

10 years building and shipping apps from scratch: SwiftUI and UIKit on iPhone, iPad, and Mac, Android apps on Google Play, plus the networking layers and device-side services those apps talk to. Most recent work is an all-new SwiftUI app for professional camera crews, running against real-time firmware, now installed with nearly every camera in the fleet.

SKILLS

Swift, SwiftUI, UIKit, Metal, Swift Concurrency, AVFoundation, Core ML, Mac Catalyst, Swift-C/C++ interop, Java (Android); WebSocket and HTTP APIs, FlatBuffers, UDP video streaming, network and serial device comms; C++, C, Python, Figma

WORK EXPERIENCE

Freefly Systems, Inc. — Woodinville, WA (remote since 2020)
Staff Software Engineer, January 2024 – Present
Senior Software Engineer, March 2021 – January 2024
Software Engineer, May 2018 – March 2021
Product Pages at https://www.freeflysystems.com

Ember Camera App — *Live preview, playback, and camera control for a camera shooting 5K at 600 FPS and 4K at 900 FPS. Used by National Geographic to film the Artemis II launch at 2,000 FPS, and on Red Bull Media House's Apple Immersive film World of Red Bull: Backcountry Skiing.*

- Architected an all-new SwiftUI companion app from scratch, wrote nearly all of the tickets the team built it from, led the app team through three major releases, and owned several customer-facing features end to end
- Designed the app's state layer, keeping camera state independent of the SwiftUI views that render it
- Shipped the app to approximately 1,500 installs against a fleet of approximately 1,600 cameras, reaching near-universal adoption among professional and scientific users; available on iOS, macOS and iPadOS
- Implemented live preview and image analysis tools with Metal, and hand-ported 'Amplify' motion amplification from DX12 to Metal, bringing it from Wave Player into the next-generation Ember Camera app
- Ported IMU-based stabilization into the app and built the user-facing tooling around it, shipping a research algorithm as a production feature
- Added tests for the app's handling of malformed clips from the camera: clips cut off by power loss with only a filename, older clips missing fields, and intact clips whose thumbnails fail to arrive
- Covered product and design work directly when those roles were unavailable, taking features from idea to production UI

Ember Camera Platform — *Camera-side services and APIs behind every Ember client: control, recording, and offload over Wi-Fi, ethernet, and USB*

- Own the code running on the Linux half of the camera and the full comms layer between it and our applications
- Defined and built the camera's private HTTP control API
- Built and maintained the WebSocket API for app-to-camera control, carrying it through a migration from a mailbox system to Google FlatBuffers, chosen for stable message sizing and schema evolution
- Architected Wi-Fi, ethernet, and USB comms between real-time firmware and client apps using portable C++ middleware
- Extended the platform to a second camera, the Ember S2.5K: new sensor with higher FPS - same architecture, same app

Movi Gimbal Apps — *Companion apps for Freefly's gimbal line, from the Movi phone gimbal to the Movi Pro, Carbon, and XL*

- Wrote the Movi Android app above its Bluetooth LE layer, including onboarding, the UI, and the camera and video pipeline, and managed release, testing, and development schedule
- Built the Android app's gimbal connection interface on an existing Bluetooth LE library, including nearest-device selection by signal strength and Android's location-permission rules for BLE scanning
- Shipped gimbal firmware updates through the iOS and Android apps across the Movi line, bundling verified firmware packages with updated checksums into app releases and keeping the Android apps current with Google Play requirements

Astro and Alta X Ground Control — *Android ground control software used by operators of Freefly's various aerial platforms*

- Integrated our flight application into the vendor-provided Herelink Android environment and shipped tested release images, so operators updated a validated system rather than dropping a new binary into an untested OS; deployed on hundreds of customer aircraft still flying today

Wave Player App — *Convert Wave Camera Raw to H.264, HEVC, or ProRes with tools to adjust color and apply LUTs*

- Owned the project end to end as sole application engineer: built the repo, published it to the Mac and iPad App Stores
- Wrote the color pipeline managing color and adjustments on Wave Raw clips with Metal
- Integrated a trained CNN upscaler into the macOS app, correcting subpixel artifacts in 2K capture and reconstructing to 4K
- Built one-click color profile and export queue systems to transform user workflows

Honeyfire, Inc. — Tacoma, WA
App Developer, May 2016 – May 2018
Project Photos and Videos: https://www.banj.io

The Myngo Project — *A Tacoma, Washington-centric delivery service for eclectic eats and everyday goods.*

- Designed and architected the Myngo and MyngoDrive Android applications from zero to launch on Google Play
- Integrated in-app payments through PayPal's Braintree API and push notifications through Firebase

Flypost Universalis — *An events platform for university students to share and discover what is happening on their campus and in their community. Available on a per-school basis for web, Android, and iOS.*

- Designed and architected Flypost for Android phones, Android TV, and Android Wear, previously published on Google Play
- Worked with university student govt. to ingest campus events from various clubs and departments, and student govt. media

EDUCATION

Bachelor of Science, Computer Science, Math Minor May 2018
University of Puget Sound, Tacoma, WA
Capstone in Computer Science January 2018 – May 2018
Research goal *Achieve autonomous drone control with an external computer vision system*

- Tracked and maintained 3D position of the drone using an Xbox 360 Kinect camera system and our software
- Designed and implemented a communications protocol between drone control and vision processing computers

MILDLY AMUSING FACT

Another software engineer friend and I maintain Scribe, a lightweight server that sends messages to the receipt printers in our homes so we can write to each other whenever. Scribe's tagline is simply "Write that down!"